

SNAPSHOT AI

Technical Documentation

Implementation notes for the competition submission.

Architecture

Input → local preprocessing → TF-IDF vectorization → Logistic Regression → security heuristics → explainable risk output.

Repository Structure

src/app.py — desktop UI and inference logic

data/messages.csv — demonstration training data

docs/ARCHITECTURE.md — system design

docs/DEMO_SCRIPT.md — live demonstration steps

README.md — setup and project overview

Local Security Signals

The prototype checks suspicious language, insecure HTTP links, raw IP URLs, unusually long URLs, @ symbols in URLs, repeated urgency and requests for OTP/PIN/CVV/card information.

Snapdragon Optimization Boundary

The model is deliberately isolated from the UI. That allows a future ONNX representation to replace the current Python inference path without redesigning the product interface. The intended target is a Snapdragon-powered Windows PC.

Testing Philosophy

The project separates demonstration behavior from production claims. A production security detector needs a larger licensed corpus, cross-domain validation, adversarial testing, false-positive analysis and device profiling.